

SIMATS ENGINEERING

CO4 - ASSESSMENT TOOL 2

Industry Problem Solving Task

COURSE NAME: Deep Learning

COURSE CODE: MLA0408

COURSE OUTCOME COVERED

CO 1: Students will be able to understand and apply probabilistic graphical models including Bayesian Networks, Markov Random Fields, Hidden Markov Models, and Conditional Random Fields. They will learn how to represent complex probabilistic relationships among variables and perform inference and learning in graphical models. Students will be able to use these models for reasoning under uncertainty and solving real-world decision-making problems. (BL-6)

Assessment Tool Weightage: 40%

QUESTIONS:

Total Marks:25

Q1. Transfer Learning

A healthcare startup aims to develop an AI-based system for detecting skin cancer from dermoscopic images. Since only a limited number of labeled images are available, design a transfer learning strategy using a pre-trained CNN model. Justify your choice of model, identify which layers should be frozen or fine-tuned, and explain how your design improves classification performance.

Q2. DenseNet and PixelNet

A smart city project requires an automated road-scene analysis system that accurately identifies roads, pedestrians, vehicles, and traffic signs at the pixel level. Design a deep learning architecture using DenseNet and PixelNet to solve this problem. Explain how the proposed architecture improves feature reuse, segmentation accuracy, and computational efficiency.

Q3. Bidirectional RNN and Encoder–Decoder

A multinational customer service company wants to build a multilingual chatbot capable of translating customer queries while preserving contextual meaning. Design an Encoder–Decoder sequence-to-sequence model using Bidirectional RNNs. Illustrate the architecture and justify how bidirectional processing improves translation accuracy.

Q4. BPTT and LSTM

A financial analytics company is developing a model to forecast stock prices using historical market data. Create an LSTM-based prediction system and explain how Backpropagation Through Time (BPTT) is used to train the network. Justify why LSTM is more suitable than a traditional RNN for this application.

Q5. Integrated Deep Learning Solution

A media streaming company plans to automatically generate captions for uploaded videos to improve accessibility. Design an integrated deep learning solution that combines Transfer Learning for visual feature extraction with an LSTM-based Encoder–Decoder architecture for

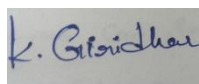
caption generation. Explain the role of each component and justify how the complete system produces accurate captions.

Code of Conduct:

I K.GIRIDHAR-192425160 certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:



RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Industry Problem	20%	Comprehensive understanding of real-world problem and constraints	Good understanding with minor gaps in requirements	Basic understanding; some requirements unclear	Poor understanding; misinterprets problem context
Application of Engineering Concepts	25%	Applies advanced and relevant concepts effectively to solve the problem	Applies appropriate concepts with minor errors	Limited application of concepts	Incorrect or no application of concepts
Solution Design	25%	Innovative, practical, and feasible solution aligned with industry standards	Logical and feasible solution with minor gaps	Basic solution with limited feasibility	Unclear or impractical solution
Use of Tools	15%	Effective use of modern tools, technologies, and real data	Appropriate use of tools with correct implementation	Limited or inconsistent use of tools	Minimal or incorrect use of tools
Analysis	10%	Thorough analysis with validated results and performance evaluation	Good analysis with reasonable validation	Basic analysis with limited validation	Poor or no analysis

Q1. Transfer Learning – Skin Cancer Detection:

1. Problem Understanding:

A healthcare startup wants to detect **skin cancer from dermoscopic images**. The major challenge is that only a limited number of labeled medical images are available.

Training a CNN completely from scratch would require a large labeled dataset. Therefore, a **pre-trained CNN using Transfer Learning** is appropriate.

2. Proposed Solution:

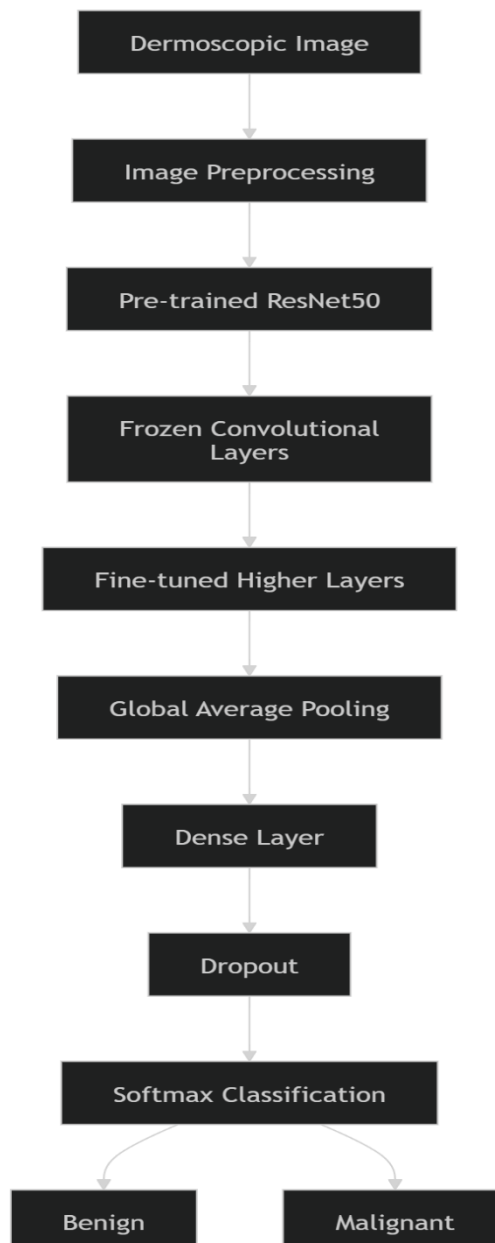
Use **ResNet50**, pre-trained on ImageNet.

The pre-trained network already contains useful low-level and mid-level features such as:

- Edges
- Textures
- Shapes
- Patterns

These features can be transferred to dermoscopic images.

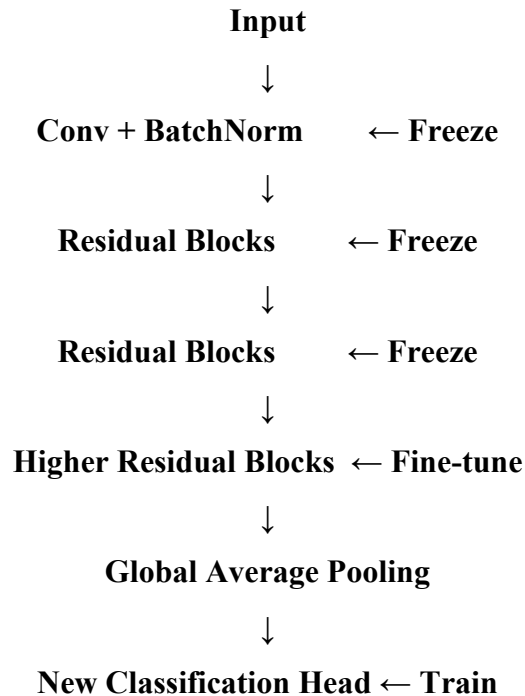
3. Architecture:



4. Frozen and Fine-Tuned Layers:

Initially, freeze most of the ResNet50 convolutional layers.

ResNet50



The first layers learn generic features, while the deeper layers learn more task-specific features.

5. Training Strategy:

Step 1: Load ResNet50 with ImageNet weights.

Step 2: Remove the original ImageNet classification layer.

Step 3: Add:

- Global Average Pooling
- Dense layer
- Dropout
- Binary/Softmax output layer

Step 4: Freeze the convolutional base.

Step 5: Train the new classification layers.

Step 6: Unfreeze the final few ResNet blocks.

Step 7: Fine-tune using a small learning rate.

6. Why Transfer Learning?

Transfer learning is suitable because:

- The available medical dataset is small.
- Pre-trained features reduce training requirements.
- Training becomes faster.
- It reduces overfitting.

- The model can achieve better classification performance than training from scratch.

7. Performance Evaluation:

Use:

- Accuracy
- Precision
- Recall
- F1-score
- Confusion Matrix
- ROC-AUC

For healthcare, **recall/sensitivity is especially important**, because missing a malignant case can be serious.

8. Result:

The proposed ResNet50 transfer-learning model can efficiently classify dermoscopic images into **benign and malignant categories**, while requiring considerably less training data than a CNN trained from scratch.

Q2. DenseNet and PixelNet – Road Scene Analysis:

1. Problem Understanding:

A smart-city project requires an automated road-scene analysis system.

The system must identify different objects **at the pixel level**:

- Road
- Pedestrians
- Vehicles
- Traffic signs

Therefore, the system requires accurate **pixel-level segmentation**.

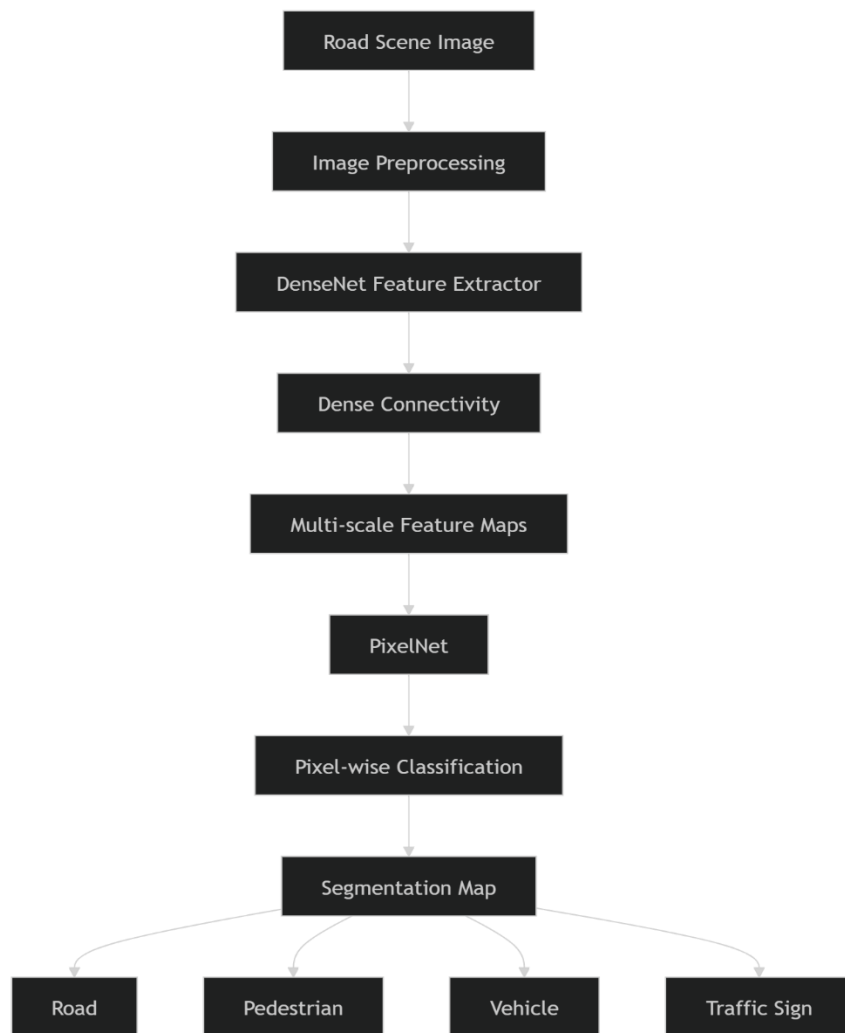
2. Proposed Solution:

The proposed architecture combines:

- **DenseNet** for feature extraction
- **PixelNet** for pixel-level classification

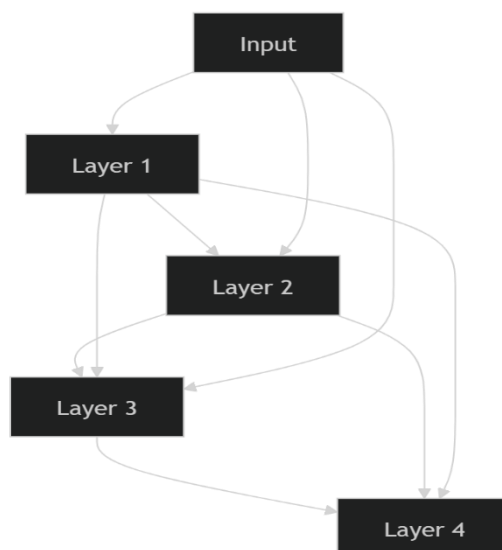
DenseNet provides efficient feature reuse through dense connections, while PixelNet performs classification at the pixel level.

3. Architecture:



4. DenseNet Feature Reuse:

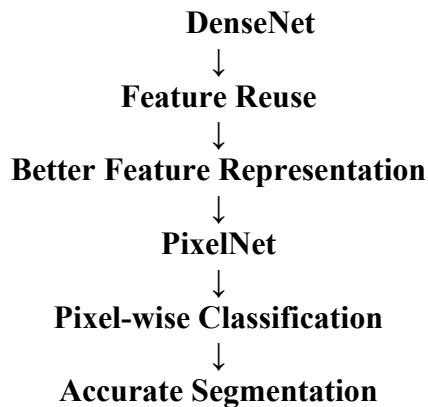
DenseNet connects each layer with the subsequent layers so that previously learned features can be reused.



This allows the network to reuse previously extracted features instead of learning the same features repeatedly.

5. PixelNet Pixel-Level Classification

PixelNet receives the extracted features and predicts the class of individual pixels.



6. How the Architecture Improves Performance:

Feature Reuse-

DenseNet's dense connections allow features from earlier layers to be reused by later layers.

Segmentation Accuracy-

PixelNet performs **pixel-wise classification**, allowing individual pixels to be assigned to the correct road-scene category.

Computational Efficiency-

DenseNet reduces redundant feature learning through feature reuse, making the architecture more parameter-efficient.

7. Performance Evaluation:

The system can be evaluated using:

Metric	Purpose
Pixel Accuracy	Measures correctly classified pixels
Mean IoU	Measures overlap between predicted and actual regions
Precision	Measures correctness of predictions
Recall	Measures detected target pixels
F1-Score	Balances precision and recall

8. Final Result:

The proposed **DenseNet + PixelNet architecture** provides pixel-level road-scene analysis. DenseNet performs efficient feature extraction and feature reuse, while PixelNet produces accurate pixel-wise segmentation for **roads, pedestrians, vehicles, and traffic signs**.

Q3. Bidirectional RNN and Encoder–Decoder:

1. Problem Understanding:

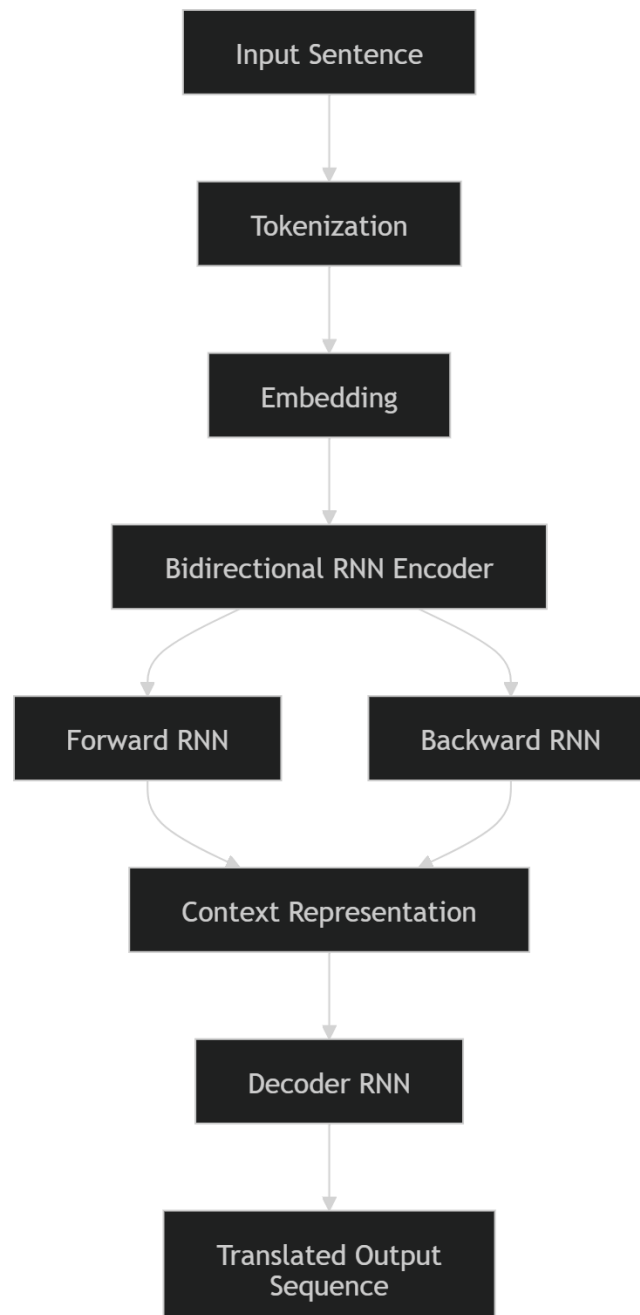
A multinational customer service company wants a **multilingual chatbot** capable of translating customer queries from one language to another while preserving the context and meaning of the original sentence.

A sequence-to-sequence **Encoder–Decoder architecture with Bidirectional RNN** is proposed for this problem.

2. Proposed Architecture:

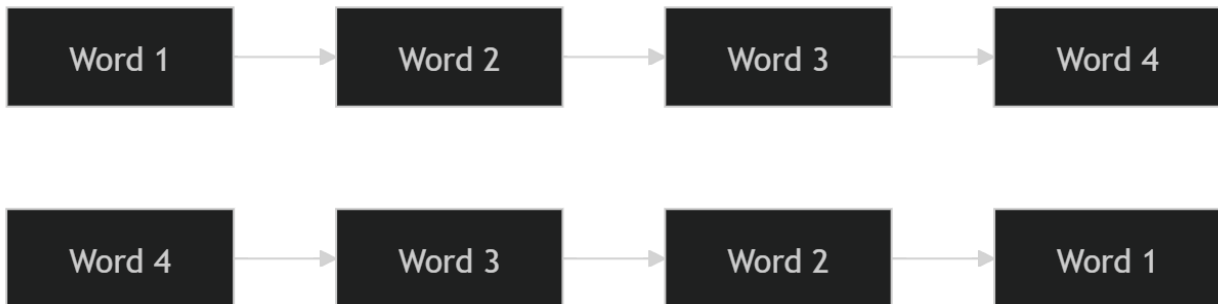
The **Bidirectional RNN Encoder** processes the input sentence in both forward and backward directions. The resulting contextual information is passed to the decoder, which generates the translated sentence.

3. Architecture:



4. Bidirectional Processing:

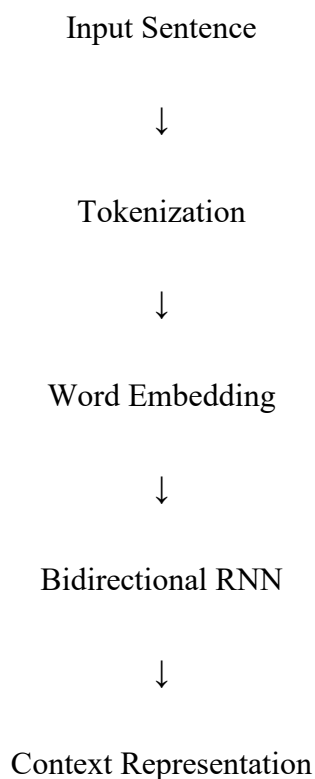
A normal RNN processes the sequence in one direction, whereas a Bidirectional RNN processes the input in both directions.



The forward RNN captures information from the beginning to the end of the sentence, while the backward RNN captures information from the end to the beginning.

5. Encoder:

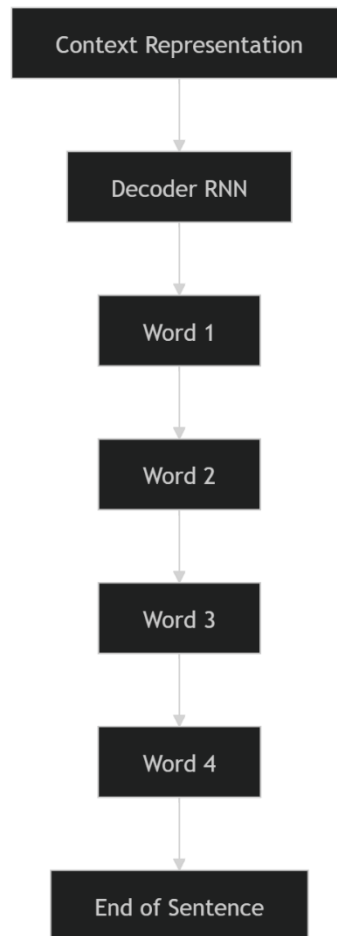
The encoder converts the input sentence into a contextual representation.



The forward and backward hidden states are combined to provide richer contextual information.

6. Decoder:

The decoder uses the context representation to generate the translated sentence sequentially.



7. Why Bidirectional Processing Improves Translation:

Bidirectional processing improves translation because the model can use information from **both the preceding and following words** when understanding the input sentence.

This helps the model:

- Understand sentence context.
- Handle word dependencies.
- Resolve ambiguous meanings.
- Generate more accurate translations.
- Preserve the meaning of customer queries.

8. Final Result:

The proposed **Bidirectional RNN Encoder–Decoder** processes the input sentence in both directions to obtain richer contextual information. The decoder then uses this representation to generate the translated sequence, improving translation accuracy and contextual understanding.

Q4. BPTT and LSTM – Stock Price Prediction:

1. Problem Understanding:

A financial analytics company wants to predict future stock prices using historical market data. Since stock prices form a time sequence, the model must learn relationships between past and future observations.

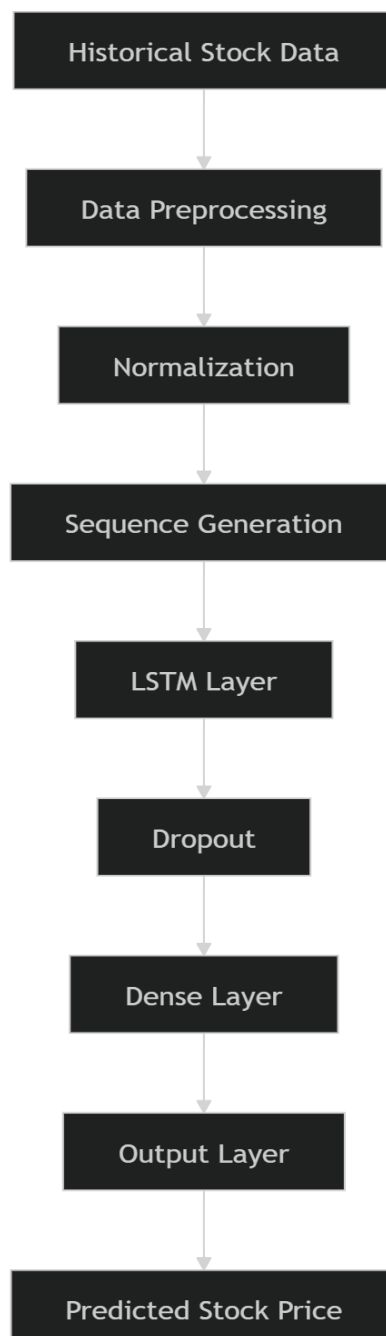
An **LSTM network** is suitable because it can learn dependencies across different time steps.

2. Proposed Solution:

The proposed system takes historical stock data as input, preprocesses and converts it into sequences, and then feeds these sequences into an LSTM network.

The LSTM learns temporal patterns and produces the predicted stock price.

3. Architecture:

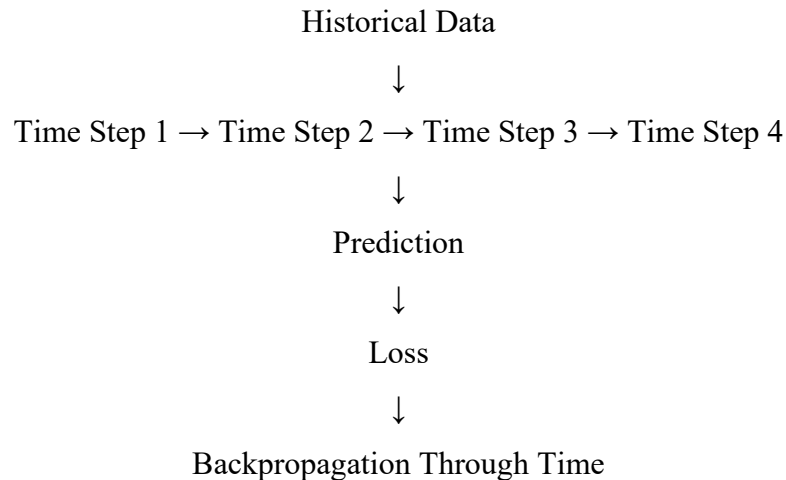


4. BPTT – Backpropagation Through Time:

Backpropagation Through Time (BPTT) is used to train the recurrent network.

The LSTM is considered across multiple time steps. After producing a prediction, the error is calculated. This error is propagated backward through the previous time steps, and the network weights are updated.

For example:



BPTT allows the model to learn how earlier stock-price information contributes to the final prediction.

5. Why LSTM is Better Than Traditional RNN:

A traditional RNN can have difficulty learning long-term dependencies because of the **vanishing-gradient problem**.

LSTM addresses this using its memory mechanism and gates:

- **Forget Gate:** decides which information should be removed.
- **Input Gate:** decides which new information should be stored.
- **Output Gate:** controls the information passed to the next state.
- **Cell State:** carries important information across time steps.

Therefore, LSTM is more suitable for stock-price forecasting where information from earlier time periods may influence future values.

6. Prediction Process:

1. Collect historical market data.
2. Preprocess the data.
3. Normalize the values.
4. Create sequences of historical observations.
5. Feed the sequences into the LSTM.
6. Train the model using BPTT.
7. Generate the predicted stock price.
8. Compare predicted and actual prices.

7. Performance Evaluation:

The prediction system can be evaluated using:

- **MAE** – Mean Absolute Error
- **MSE** – Mean Squared Error
- **RMSE** – Root Mean Squared Error
- **R² Score** – Measures the goodness of fit

8. Final Result:

The proposed **LSTM-based stock-price prediction system** learns temporal relationships from historical market data. BPTT allows the network to update its parameters based on errors across different time steps, while LSTM's memory and gating mechanisms make it more suitable than a traditional RNN for learning long-term dependencies.

Q5. Integrated Deep Learning Solution:

1. Problem Understanding:

A media streaming company wants to automatically generate captions for uploaded videos to improve accessibility.

The system must understand the visual content of the video and convert that visual information into meaningful natural-language captions.

2. Proposed Solution:

The proposed solution combines two major components:

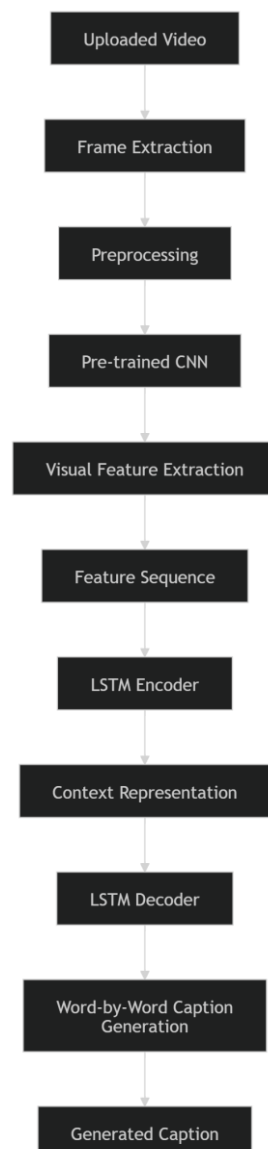
Transfer Learning:

A pre-trained CNN extracts important visual features from video frames.

LSTM Encoder–Decoder:

The LSTM encoder learns the sequence of visual features, while the LSTM decoder generates the caption word by word.

3. Architecture:



4. Transfer Learning Component:

A pre-trained CNN is used to extract visual information from each video frame.

Instead of training the CNN from the beginning, the knowledge learned from a large image dataset is reused.

The CNN extracts features such as:

- Objects
- Shapes
- Textures
- Visual patterns
- Scene information

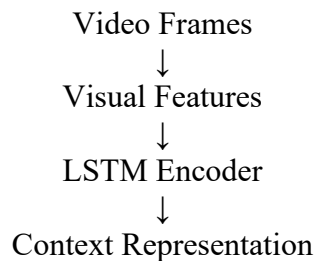
These features are then passed to the LSTM encoder.

5. LSTM Encoder:

The extracted visual features from consecutive video frames are provided to the LSTM encoder.

The encoder learns the temporal relationship between the frames and creates a **context representation** of the video.

For example:

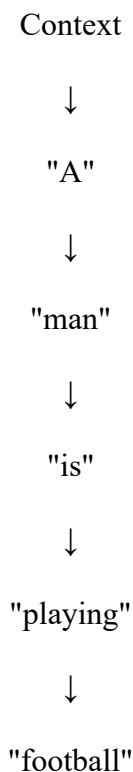


6. LSTM Decoder:

The decoder uses the context representation to generate the caption sequentially.

It predicts one word at a time until the complete caption is produced.

For example:



The resulting caption could be:

"A man is playing football."

7. Role of Each Component:

Component	Role
Video	Input source
Frame Extraction	Converts video into individual frames
Pre-trained CNN	Extracts visual features
Transfer Learning	Reuses previously learned visual knowledge
LSTM Encoder	Learns temporal information
Context Representation	Represents the video content
LSTM Decoder	Generates the caption
Output	Final generated caption

8. Why the Complete System Works?

The two techniques perform different tasks.

Transfer Learning provides strong visual feature extraction without training a CNN completely from scratch.

LSTM Encoder understands the sequence of visual information from the video.

LSTM Decoder converts the learned representation into natural-language text.

Therefore:

Video → Visual Features → Temporal Understanding → Language Generation → Caption

9. Performance Evaluation:

The generated captions can be evaluated using:

- **BLEU**
- **ROUGE**
- **METEOR**
- **CIDEr**

These metrics compare the generated captions with reference captions.

10. Final Result:

The proposed integrated system combines **Transfer Learning and LSTM Encoder–Decoder architecture** to automatically generate meaningful captions from uploaded videos. The CNN extracts visual information, while the LSTM components learn the sequence and generate the final caption, providing an effective solution for video accessibility